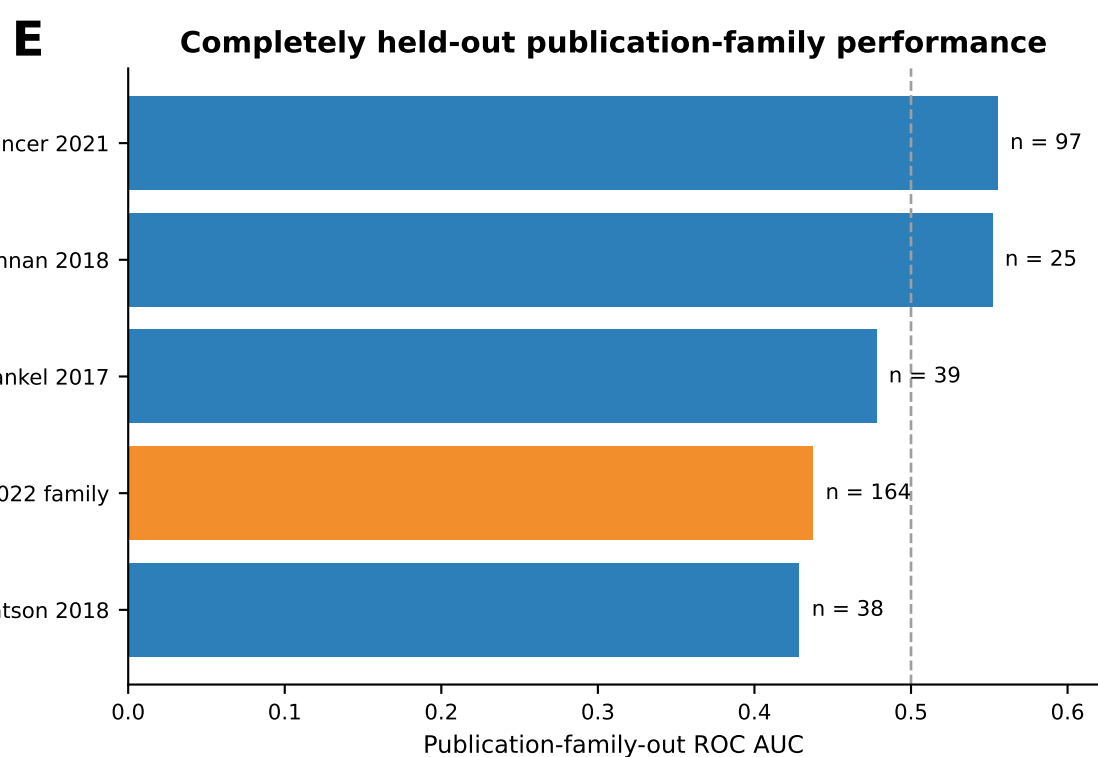
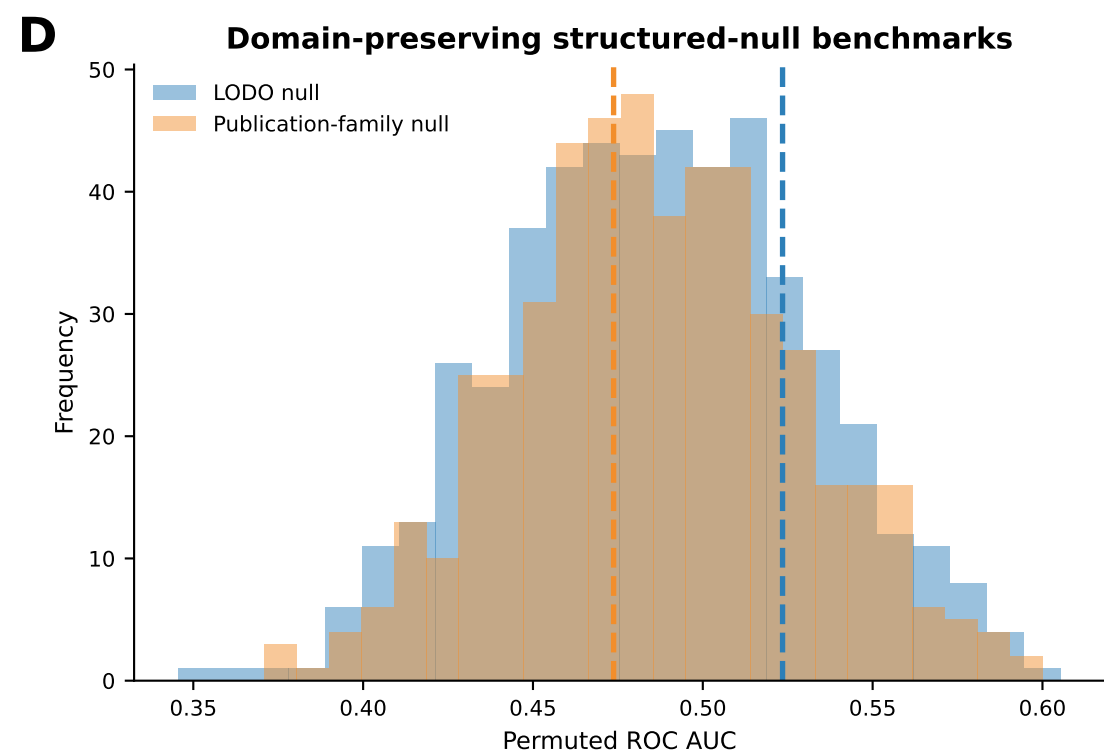
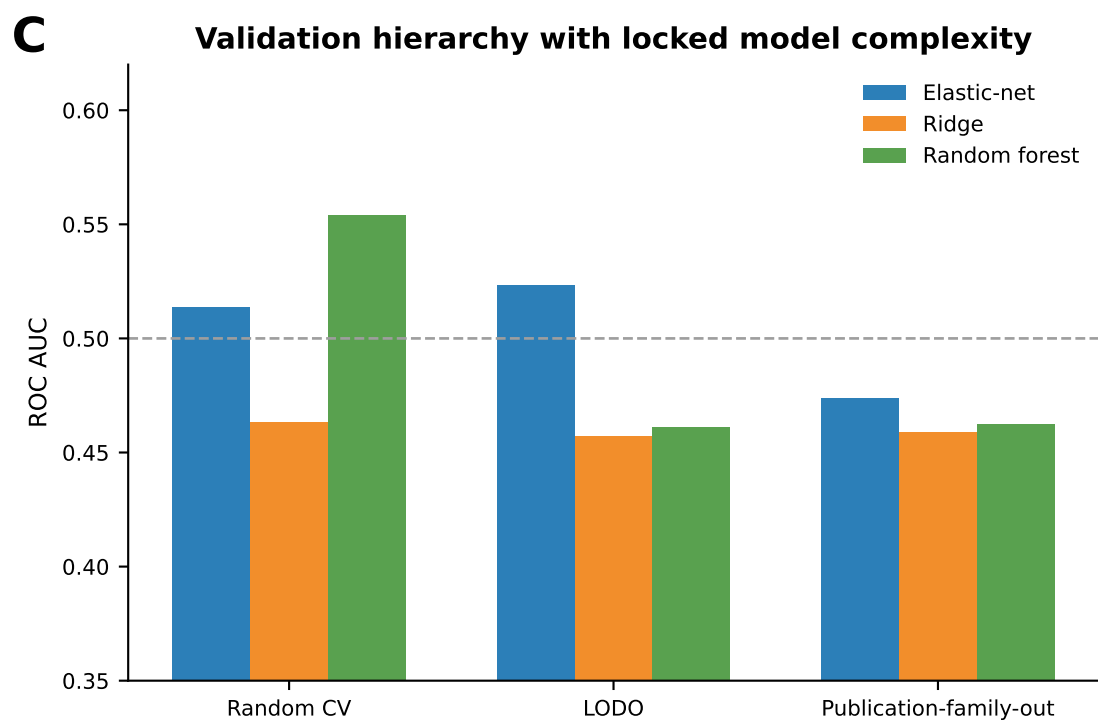
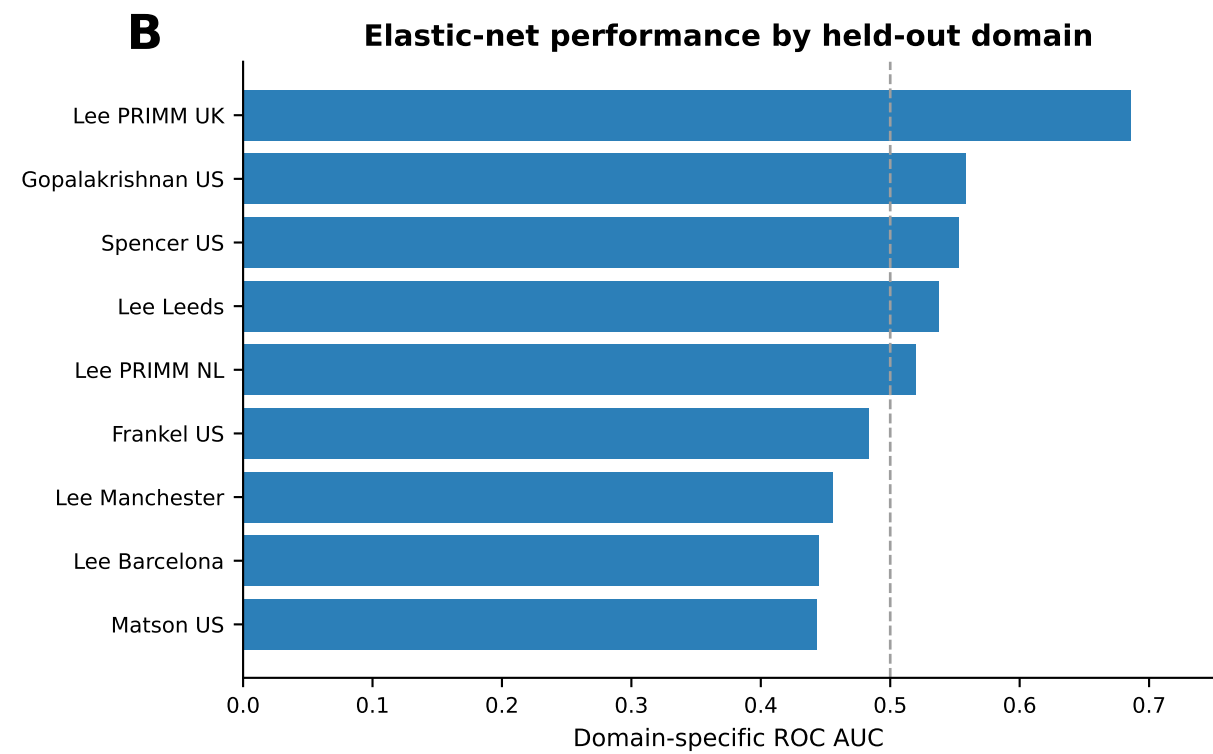
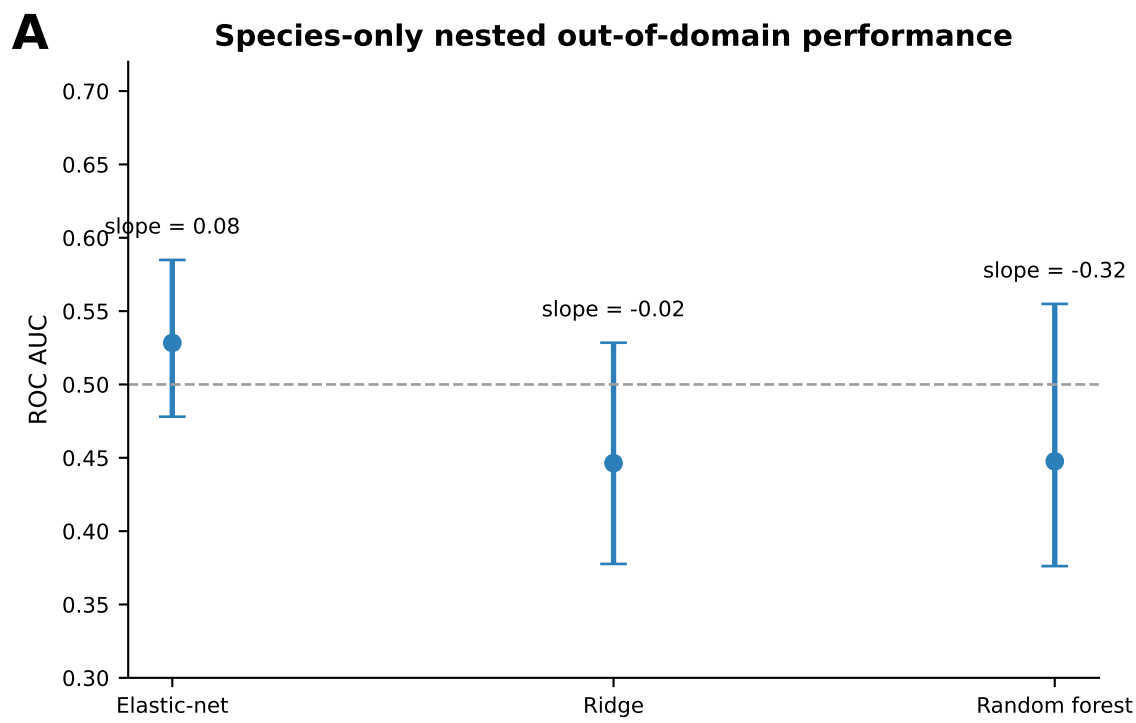


Species-level prediction performance and validation hierarchy



F Prespecified interpretation of the validation hierarchy

Structured-null result

Fixed LODO: ROC AUC = 0.523, P = 0.216. Publication-family-out: ROC AUC = 0.474, P = 0.607.

Partition sensitivity

The pooled 0.050 difference was concentrated in the five-site Lee family; the other four publication families each contained one domain and therefore had identical partitions.

Inference boundary

With only five publication families and one multi-domain family, the data do not establish a general publication-family penalty.